

CN MID-2 IMPORTANT TOPIC LIST

SR	TOPIC NAME	SUB-TOPIC NAME	Key Terms	Details
1	IPv6/IPng	IPv6 Features	Features of IPv6	Larger Address Space, Simplified Header, End-to-end Connectivity, Auto-configuration, Faster Forwarding/Routing, IPSec, No Broadcast, Anycast Support, Mobility, Enhanced Priority Support, Smooth Transition, Extensibility
2		IPv6 Addressing Modes	Anycast, Unicast, Multicast	Diagram and Meaning
3		IPv6 Addressing	Format of Address/ Address Structure	128 bits arrangement in different blocks
			Rules to Shorten IPv6 Address	How to compress and uncompress IPv6 Address
			Validity of IPv6 Address	Find whether given IPv6 Address is valid or not
4		IPv6 Address Types	Interface ID, Global Unicast Address, Link Local Address, Unique Local Address	Structure, Meaning, Compare with IPv4, Scope Diagram

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5	Transmission Media	Types of TM	Wired vs Wireless	Difference and Classification Diagram
6		Guided Media	List out	
7		Twisted Pair Cable	Why Twist?	
			Types of TPC: STP vs UTP	Structure, Difference
			Standards: CAT	CAT 1 to till Date with Data rate and Applications
			Connectors	List, Full Form and Application
			Applications	
			Advantages and Disadvantages	
			8	Co-axial Cable
Structure				
Standards: RG				
Connectors: BNC				
Applications				
Advantages and Disadvantages				
9		Fiber Optic Cable		
			Propagation Modes: Single Mode, Multi Mode, Step Index and Graded Index	Classification Diagram and details of each with diagram
			Sources of Light	LASER and LED
			Structure	
			Connectors	
			Applications	
			Advantages and Disadvantages	
10		Unguided Media	List out	
			EMS	Ranges of Frequency (VLF to EHF)
			Propagation Methods: Ground, Sky, Line of Sight	Diagram and Details
			Radio wave Communication	Range of Frequency, Signal Characteristics, Types of Antennas, Applications
			Micro wave Communication	Range of Frequency, Signal Characteristics, Types of Antennas, Applications
			Infrared Communication	Range of Frequency, Signal Characteristics, Types of Antennas, Applications
			LASER Communication	
	Satellite Communication		Components, Uplink, Downlink, Geosynchronous Satellite Applications	

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11	Network Devices	List out Devices	Layerwise Classify Network Devices	
12		NIC	Hardware Address and Details	Address Structure, Application and Details
13		Repeater	Working, Types	Why Repeater is Not an Amplifier? Diagram and Details Type wise
14		Hub	Working, Types	Diagram and Details Type wise
15		Bridge	Working, Types	Diagram and Details Type wise
16		Switch: Layer-2	Working, Types	Diagram and Details Type wise, Port Status, Switching Methods: Store and Forward, Cut Through, and Fragment Free
17		Switch: Layer-3	Working, Types	Diagram and Details Type wise: VLAN, Routing
18		Router	Working, Types	Diagram and Details Type wise: Routing, IP Addressing
19		Gateway (Protocol Converter)	Working, Types	Diagram and Details Type wise
20		Access Point	Working, Types	Diagram and Details: BSS, ESS
21		Differences	Between any Two Devices Listed above	

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22	Network Security Aspects	Basics of Security	CIA Triad	Diagram and Terms
23		Firewall	Design Principles	Diagram, Details and Applications
			Types of Firewall: Hardware vs Software Firewall, Packet Filtering Firewall, Statefull Inspection Firewall, Circuit Level Firewall, Application (Proxy) Firewall, Next Generation Firewall	
24		Security Threats	Types of Threats	Virus, Worms, Trojan, Back door, Trap Door, Malwares, Intruders, Insiders, Criminal organizations, Terrorists, Information warfare